Experiment: 2

PART B

(PART A: TO BE COMPLETED AND SUBMITTED BY STUDENTS)

Students must execute all the programs, write executed code in the workbook, and submit part B of experiment 1 on the student portal. The filename should be OOPL_batch_rollno_experimentno. Example: OOPL_A1_E001_E2

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Date of Experiment: 12/08/2026	**Date of Submission: 12/08/2026**

Aim: Implement a program using Encapsulation, Abstraction, and an Array of Objects.

Problem Statement:

A school is developing a library management system to maintain the records of the books. Where it performs basic operations such as making an entry in the book, searching the book by book name and author name. It also shows the availability of the books.

Tasks:

Create a class to manage the books in "Library" with the following data members(Instance Variables) and functions(methods) .

- Name of the book
- Author of the book
- Price of the book
- No. of copies

Implement member functions(methods) to perform the following operations (using an array of objects):

1. Read the data of 5 books – include books in the library.
2. Display the data of books – print the list of all the books in the library.
3. Search the specified book in the Library by name, if it is available or not, if available then display no of copies of the book.
4. Search the specified book in the Library by Author Name, display all the books of the same author.

Executed Code, Input and Output

1.	Implement a program using Encapsulation, Abstraction, and an Array of Objects.

Executed Code: -

```
import java.util.*;


class Book {

   String bookName;

   String author;

   double price;

   int copies;


   public void readBook() {

      Scanner sc = new Scanner(System.in);


      System.out.println("Enter book name:");

      bookName = sc.nextLine();


      System.out.println("Enter author:");

      author = sc.nextLine();


      System.out.println("Enter book price:");

      price = sc.nextDouble();


      System.out.println("Enter no of copies:");
```

```
        copies = sc.nextInt();

        sc.nextLine();

    }


    public int getCopies() {

        return copies;

    }


    public boolean searchBook(String name) {

        return bookName.equalsIgnoreCase(name);

    }


    // Search by author

    public boolean searchByAuthor(String authorName) {

        return author.equalsIgnoreCase(authorName);

    }


    public void displayBook() {

        System.out.println("Book Name: " + bookName);

        System.out.println("Author: " + author);

        System.out.println("Price: " + price);

        System.out.println("No of Copies: " + getCopies());

    }
```

```
}


public class LibraryManagement {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.println("Enter number of books:");
      int n = sc.nextInt();
      sc.nextLine();


      Book[] books = new Book[n];


      // Input book details
      for (int i = 0; i < n; i++) {
         books[i] = new Book();


         System.out.println("\nEnter details of book " + (i + 1));
         books[i].readBook();
      }


      // Display book details
      System.out.println("\nBook Details:");
```

```
for (int i = 0; i < n; i++) {
    System.out.println("\nDetails of book " + (i + 1));
    books[i].displayBook();
}

// Search by book name
System.out.println("\nEnter book name to search:");
String name = sc.nextLine();

boolean found
```

Input Output: -

```
Author: MK Gandhi
Price: 800.0
No of Copies: 10

Details of book 2
Book Name: Mein Kamf
Author: A. Hitler
Price: 500.0
No of Copies: 10

Enter book name to search:
Mein Kamf

Book found:
Book Name: Mein Kamf
Author: A. Hitler
Price: 500.0
No of Copies: 10
Book found by name: true

Enter author name to search:
A. Hitler

Book found by author:
Book Name: Mein Kamf
Author: A. Hitler
Price: 500.0
No of Copies: 10
Book found by author: true
```

Observation and Learning: -

I understood the basic concepts of OOP, especially encapsulation and abstraction. I learned that encapsulation helps protect data, while abstraction hides unnecessary details. I also understood how an array of objects can be used to manage multiple objects of the same class.

Overall, this helped me understand how OOP makes programs more organised and easier to manage.

Question of curiosity:

1. Differentiate among the following concepts: Encapsulation, Abstraction,
2. What is an Array of Objects? Explain.

1. Difference between Encapsulation and Abstraction

- **Encapsulation:** Combining data and functions into one unit and restricting direct access to the data.
- **Abstraction:** Hiding unnecessary implementation details and showing only the required features.

In simple words: Encapsulation is about **protecting data**, while abstraction is about **hiding complexity**.

2. Array of Objects

An array of objects is a collection of multiple objects belonging to the same class. Each object has its own data and can be accessed separately. It is useful for managing groups of similar objects, such as students, employees, or books.